

Transformation	${}^{i-1}\mathbf{T}_i$	${}^{i-1}\mathbf{T}_i^*$
Frame 3 to frame 4	${}^3T_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -0.0064 \\ 0 & 1 & 0 & -0.2104 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	${}^3T_4^* = \begin{bmatrix} cq_4 & -sq_4 & 0 & 0 \\ 0 & 0 & -1 & -0.0064 \\ sq_4 & cq_4 & 0 & -0.2104 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Frame 4 to frame 5	${}^4T_5 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -0.2084 \\ 0 & -1 & 0 & -0.0064 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	${}^4T_5^* = \begin{bmatrix} cq_5 & -sq_5 & 0 & 0 \\ 0 & 0 & 1 & -0.2084 \\ -sq_5 & -cq_5 & 0 & -0.0064 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Frame 5 to frame 6	${}^5T_6 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & -0.1059 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	${}^5T_6^* = \begin{bmatrix} cq_6 & -sq_6 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ sq_6 & cq_6 & 0 & -0.1059 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Frame 6 to frame 7	${}^6T_7 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -0.1059 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	${}^6T_7^* = \begin{bmatrix} cq_7 & -sq_7 & 0 & 0 \\ 0 & 0 & 1 & -0.1059 \\ -sq_7 & -cq_7 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Frame 7 to interface module	${}^7T_{INT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & -0.0615 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	



Note: units are in meters for homogeneous transform translations in the right-hand column of each matrix.

Homogeneous transform matrices - 6 DoF spherical wrist

Homogeneous transform matrices for the 6 DoF spherical wrist robot.

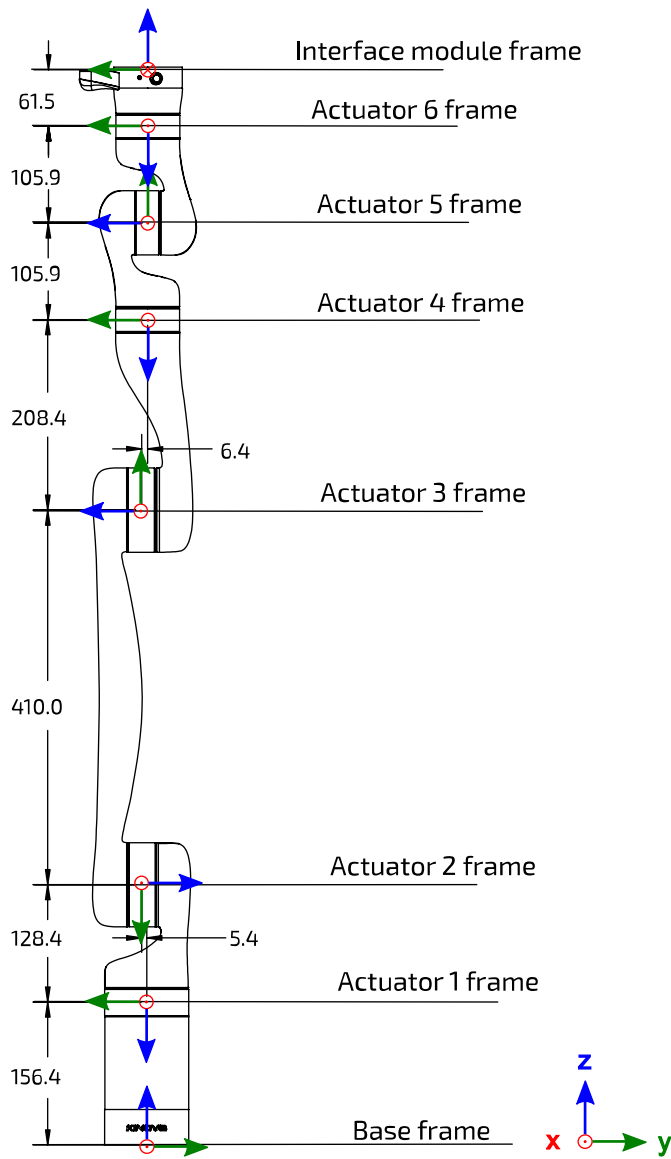


Figure 85: 6 DoF robot frame definitions and dimensions (all joints at 0 position, dimensions in mm)

Table 92: 6 DoF homogeneous transformation matrices

Transformation	${}^{i-1}\mathbf{T}_i$	${}^{i-1}\mathbf{T}_i^*$
Base to frame 1	${}^B\mathbf{T}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0.1564 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	${}^B\mathbf{T}_1^* = \begin{bmatrix} cq_1 & -sq_1 & 0 & 0 \\ -sq_1 & -cq_1 & 0 & 0 \\ 0 & 0 & -1 & 0.1564 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Frame 1 to frame 2	${}^1\mathbf{T}_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0.0054 \\ 0 & 1 & 0 & -0.1284 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	${}^1\mathbf{T}_2^* = \begin{bmatrix} cq_2 & -sq_2 & 0 & 0 \\ 0 & 0 & -1 & 0.0054 \\ sq_2 & cq_2 & 0 & -0.1284 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Transformation	${}^{i-1}\mathbf{T}_i$	${}^{i-1}\mathbf{T}_i^*$
Frame 2 to frame 3	${}^2T_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & -0.410 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	${}^2T_3^* = \begin{bmatrix} cq_3 & -sq_3 & 0 & 0 \\ -sq_3 & -cq_3 & 0 & -0.410 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Frame 3 to frame 4	${}^3T_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0.2084 \\ 0 & 1 & 0 & -0.0064 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	${}^3T_4^* = \begin{bmatrix} cq_4 & -sq_4 & 0 & 0 \\ 0 & 0 & -1 & 0.2084 \\ sq_4 & cq_4 & 0 & -0.0064 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Frame 4 to frame 5	${}^4T_5 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & -0.1059 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	${}^4T_5^* = \begin{bmatrix} cq_5 & -sq_5 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -sq_5 & -cq_5 & 0 & -0.1059 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Frame 5 to frame 6	${}^5T_6 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0.1059 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	${}^5T_6^* = \begin{bmatrix} cq_6 & -sq_6 & 0 & 0 \\ 0 & 0 & -1 & 0.1059 \\ sq_6 & cq_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Frame 6 to interface module	${}^6T_{INT} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & -0.0615 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	



Note: units are in meters for homogeneous transform translations in the right-hand column of each matrix.

Vision module sensors reference frames

Homogeneous transforms defining the reference frames of the video module sensors in terms of the interface module frame. These are useful in correlating data from the color and depth sensors.

Overview

The vision module captures data about the environment of the robot via two sensors:

- color sensor
- depth sensor

Developing vision applications and vision-based control of the robot will require correlating the data from these two sensors with each other and with the position of the end effector.

Camera frames

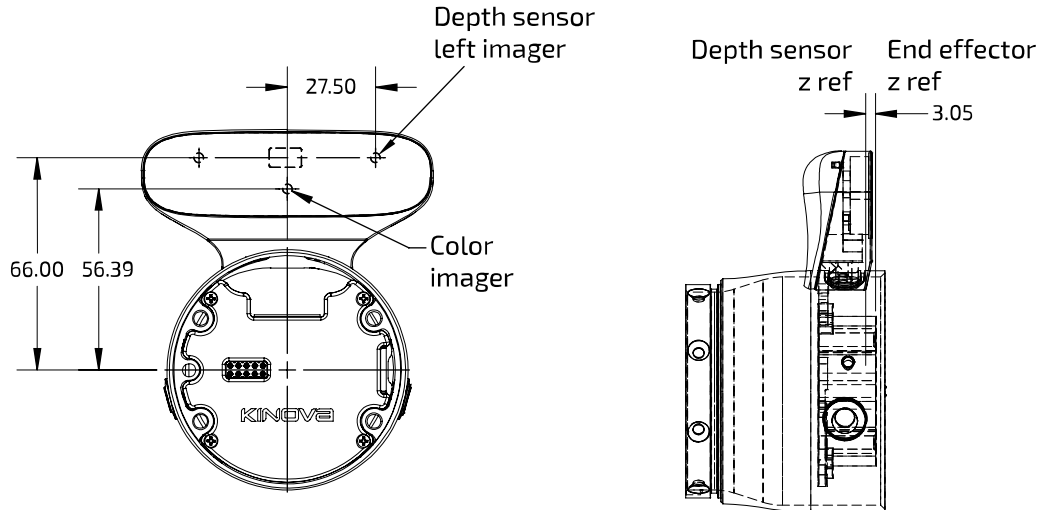


Figure 86: Vision sensors

Color sensor

The reference frame for the color sensor has the same orientation as that of the end effector, but with an offset of $(dx, dy, dz) = (0 \text{ mm}, 56.39 \text{ mm}, -3.05 \text{ mm})$.

Depth sensor

The origin for the depth center frame of reference is defined at the center of the left imager, directly in front of the lens of the left imager. The z offset for the depth sensor reference frame is the same as for the color sensor reference frame, and the y offset is 66.00 mm. Because the left imager is used as a reference, there is also an x offset of 27.50 mm.

The matrices for the reference frame transformations from end effector frame to vision sensor frame are shown in the table below.

Table 93: Reference frame transformations from end effector to sensors (displacement units in meters)

Transformation	Transformation matrix
End effector to color sensor	${}^{INT}T_{COLOR} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0.05639 \\ 0 & 0 & 1 & -0.00305 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
End effector to depth sensor	${}^{INT}T_{DEPTH} = \begin{bmatrix} 1 & 0 & 0 & 0.02750 \\ 0 & 1 & 0 & 0.06600 \\ 0 & 0 & 1 & -0.00305 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Denavit-Hartenberg (DH) parameters

DH parameters the robot are provided.

DH parameters

Denavit-Hartenberg (DH) parameters offer another convenient way to specify the reference frame transformations for the robot kinematic chain.

The Classical DH parameters convention transformation expresses a frame in terms of the previous as:

$${}^{i-1}T_i = \begin{bmatrix} \cos(\theta_i) & -\cos(\alpha_i)\sin(\theta_i) & \sin(\alpha_i)\sin(\theta_i) & a_i\cos(\theta_i) \\ \sin(\theta_i) & \cos(\alpha_i)\cos(\theta_i) & -\sin(\alpha_i)\cos(\theta_i) & a_i\sin(\theta_i) \\ 0 & \sin(\alpha_i) & \cos(\alpha_i) & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Figure 87: DH parameters Classical frames transformations

Denavit-Hartenberg (DH) parameters - 7 DoF spherical wrist

DH parameters for the 7 DoF spherical wrist robot.

Table 94: 7 DoF spherical Classical DH parameters

i	α_i (radians)	a_i (m)	d_i (m)	θ_i (radians)
0 (from base)	π	0.0	0.0	0
1	$\pi/2$	0.0	-(0.1564 + 0.1284)	q_1
2	$\pi/2$	0.0	-(0.0054 + 0.0064)	$q_2 + \pi$
3	$\pi/2$	0.0	-(0.2104 + 0.2104)	$q_3 + \pi$
4	$\pi/2$	0.0	-(0.0064 + 0.0064)	$q_4 + \pi$
5	$\pi/2$	0.0	-(0.2084 + 0.1059)	$q_5 + \pi$
6	$\pi/2$	0.0	0.0	$q_6 + \pi$
7 (to interface)	π	0.0	-(0.1059 + 0.0615)	$q_7 + \pi$

Here $q_1, q_2, \dots$ refers to the joint angles.

The figures in the chart are based on the following frames definitions:

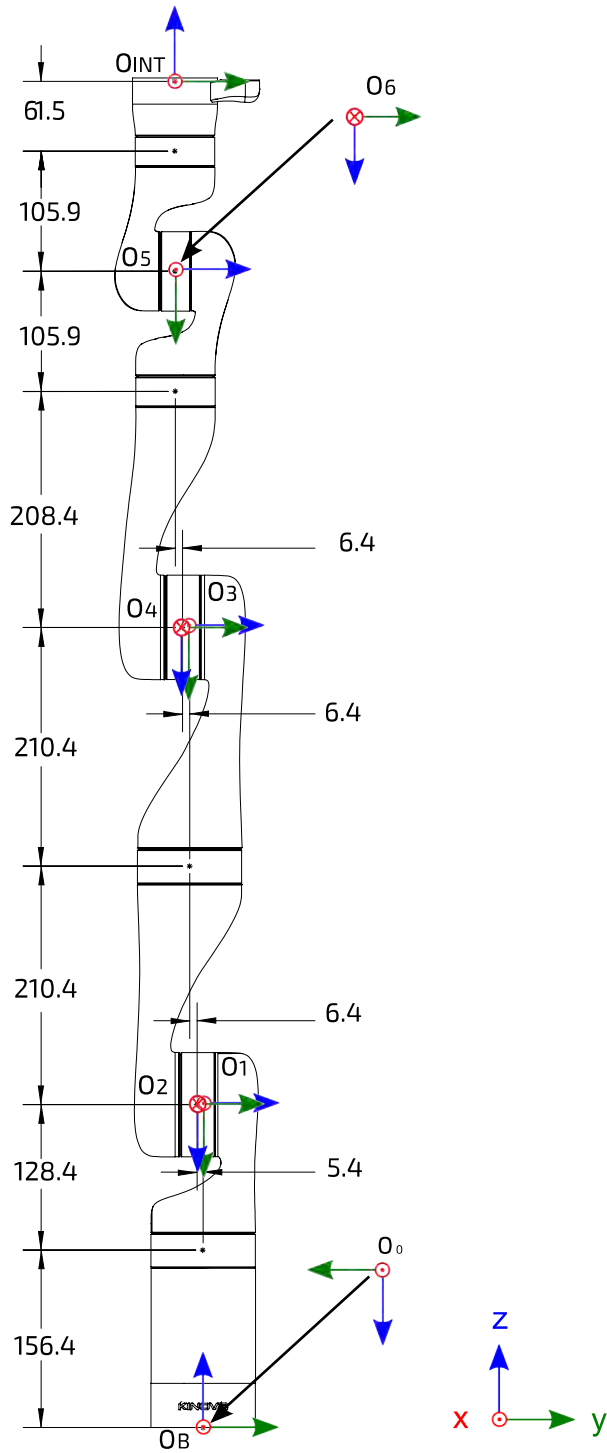


Figure 88: Denavit-Hartenberg frames definitions

Denavit-Hartenberg (DH) parameters - 6 DoF spherical wrist

DH parameters for the 6 DoF spherical wrist robot.

Table 95: 6 DoF spherical Classical DH parameters

i	α_i (radians)	a_i (m)	d_i (m)	θ_i (radians)
0 (from base)	0.0	0.0	π	0.0
1	0.0	$-(156.43 + 128.38)$	$\pi/2$	q_1
2	410.0	-5.38	π	$q_2 - \pi/2$
3	0.0	-6.38	$\pi/2$	$q_3 - \pi/2$
4	0.0	$-(208.43 + 105.93)$	$\pi/2$	$q_4 + \pi$
5	0.0	0.0	$\pi/2$	$q_5 + \pi$
6 (to interface)	0.0	$-(105.93 + 61.53)$	π	$q_6 + \pi$

Here $q_1, q_2, \dots$ refers to the joint angles.

The figures in the chart are based on the following frames definitions:

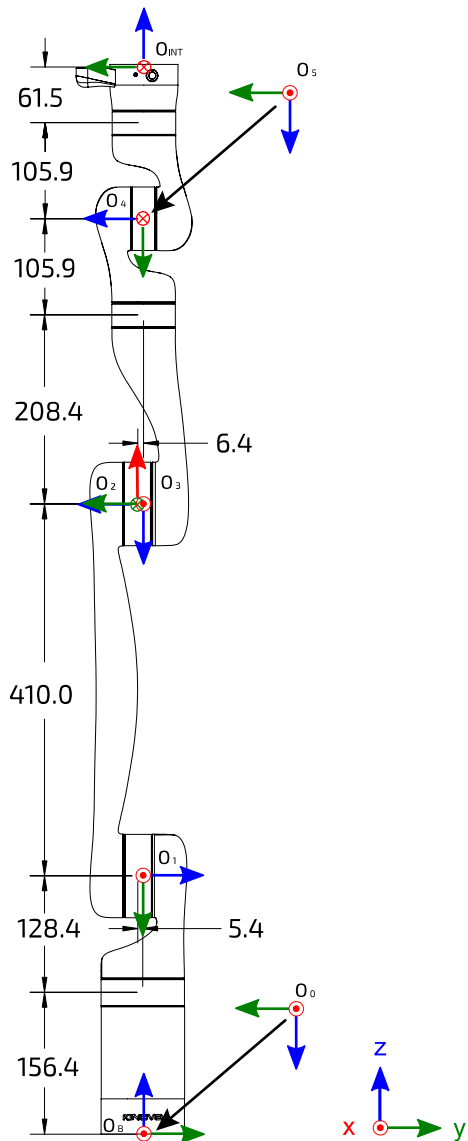


Figure 89: Denavit-Hartenberg frames definitions

7 DoF singular configurations

Description of the singular configurations of the 7 DoF robot.

Singular configurations overview

Singularities generally occur when a particular angular configuration of the robot causes axes to be aligned. This causes the robot to lose degrees of freedom and experience limitations in movement in some directions while operating the robot in Cartesian mode. There are many ways that this could potentially happen, and an exhaustive listing would be difficult. The following table highlights some important singular configurations for the 7 DoF robot, explaining how they occur and how the robot behavior is altered near the singularity while in Cartesian mode.

Table 96: Selected singular configurations description

Singularity	Definition	Description
Boundary singularity	The arm is at full reach. Joint 4 (elbow) is at 0°. The arm cannot move any farther in the direction it is currently reaching out.	The robot will slowly try to reach a straight position. If you try to move the robot purely in the x or y direction starting from this position, you will not be able to. The robot will follow the command as best as possible. It will move in x and y while going down in the z direction.
Joints 2 and 3 singularity	Joint 2 is at 0° so joints 1 and 3 are perfectly aligned and have the same effect. Joint 3 is at 90° or at 270° so that the axes of joint 2 and joint 4 are perpendicular. The robot can no longer move purely along an axis in translation.	Due to singularity crossing algorithm, the robot will make a minor deviation from a straight line trajectory when going through this pose in Cartesian mode.
Joints 2 and 6 singularity	Joint 2 is at 0° so that joints 1 and 3 are perfectly aligned and have the same effect. Joint 6 is at 0° so that joints 5 and 7 are perfectly aligned and have the same effect. The hand cannot rotate in one direction anymore.	Due to singularity crossing algorithm, the robot will make a minor deviation from a straight line trajectory when going through this pose in Cartesian mode.

Singularity	Definition	Description
Joints 5 and 6 singularity	Joint 6 is at 0° so that joints 5 and 7 are perfectly aligned and have the same effect. Joint 5 is at 90° or at 270° so that the axes of joint 4 and joint 6's axis are perpendicular. The robot can no longer complete pure rotations around an axis.	Due to singularity crossing algorithm, the robot will make a minor deviation from a straight line trajectory when going through this pose in Cartesian mode.
Joints 1 and 7 singularity	Joints 2-5 are positionned so that joints 1 and 7 are perfectly aligned and have the same effect (infinite possibilities of joint combinations).	Due to singularity crossing algorithm, the robot will make a minor deviation from a straight line trajectory when going through this pose in Cartesian mode.

6 DoF singular configurations

This section describes the singularity configurations of the 6 DoF robot.

Singularity configurations overview

Singularities generally occur when a particular angular configuration of the robot causes axes to be aligned, causing the robot to lose degrees of freedom and experience limitations in movement in some directions while operating the robot in Cartesian mode. There are many ways that this could potentially happen, and an exhaustive listing would be difficult. The following table highlights some important singularities for the 6 DoF robot, explaining how they occur and how the robot behavior is altered near the singularity while in Cartesian mode.

Table 97: Selected singular configurations description

Singularity	Definition	Description
Boundary singularity	The robot is at full extension.	The robot will slowly try to reach a straight position. If you try to move the robot purely in the x or y direction starting from this position, you will not be able to. The robot will follow the command as best as possible. It will move in x and y while going down in the z direction.
Joints 1 and 4 singularity	Joints 2 and 3 are at 0° so that joints 1 and 4 are aligned.	Due to singularity crossing algorithm, the robot will make a minor deviation from a straight line trajectory when going through this pose in Cartesian mode.

Singularity	Definition	Description
Joints 1 and 6 singularity	Joints 2, 3, and 5 are at a position such that Joints 1 and 6 are aligned and have the same effect. Happens if $\theta_2 + \theta_3 + \theta_5 = 0$ if θ is defined between $\pm \pi$.	Due to singularity crossing algorithm, the robot will make a minor deviation from a straight line trajectory when going through this pose in Cartesian mode.
Joints 4 and 6 singularity	Joint 5 is at 0° so joints 4 and 6 are perfectly aligned and have the same effect.	Due to singularity crossing algorithm, the robot will make a minor deviation from a straight line trajectory when going through this pose in Cartesian mode.

Inertial parameters definition

Inertial parameters are provided for the robot.

Overview

The following tables describe the key inertial parameters of each independently moving rigid link segment of the 7 robot. This includes:

- mass in kg
- centers of masses in meters
- moments of inertia in $\text{kg} \cdot \text{m}^2$

The six distinct moments of inertia (I_{xx} , I_{xy} , I_{xz} , I_{yy} , I_{yz} , I_{zz}) are presented in tabular form.

Conventions used

The following conventions are used:

1. The center of mass of a link is always expressed in terms of the reference frame of the precedent joint.
2. The mass of a link segment includes the shell and portions of the actuators at each end of the link (as applicable) that are enclosed within the link and move rigidly with the link.
3. The moments of inertia of the link segments are taken in a frame defined at the center of mass of the link segment and aligned with the precedent joint frame.



Note: These conventions align with those used in URDF files in the `<inertial>` section of `<link>` definitions.

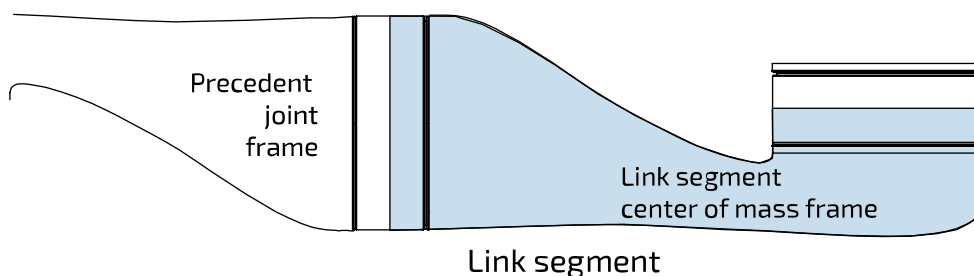


Figure 90: Inertial parameters conventions

Inertial parameters of the 7 DoF robot

Inertial parameters of the 7 DoF robot.

Overview

The following tables describe the key inertial parameters of the link segments of the 7 DoF robot.

Table 98: Base

Physical quantity	Value		
mass (kg)	1.697		
center of mass coordinates (m)	[-0.000648, -0.000166, 0.084487]		
moments of inertia (kg · m ²)	lxx	0.004622	
	lxy	0.000009	
	lxz	0.000060	
	lyy	0.004495	
	lyz	0.000009	
	lzz	0.002079	

Table 99: Link 1

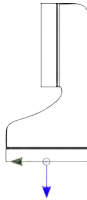
Physical quantity	Value		
mass (kg)	1.377		
center of mass coordinates (m)	[-0.000023, -0.010364, -0.073360]		
moments of inertia (kg · m ²)	Ixx	0.004570	
	Ixy	0.000001	
	Ixz	0.000002	
	Iyy	0.004831	
	Iyz	0.000448	
	Izz	0.001409	

Table 100: Link 2

Physical quantity	Value		
mass (kg)	1.1636		
center of mass coordinates (m)	[-0.000044, -0.099580, -0.013278]		
moments of inertia (kg · m ²)	Ixx	0.011088	
	Ixy	0.000005	
	Ixz	0.000000	